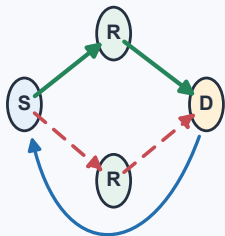


Smart-CALM: ACK-aware online redundancy control

A simple LoRa mesh data plane feeds a compact state-action controller that chooses per-flow redundancy profiles.

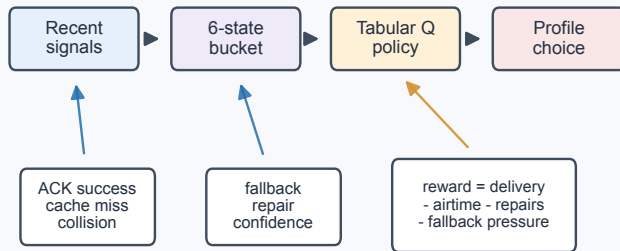
a ACK-aware forwarding plane



▶ DATA ▶ fallback ▶ ACK

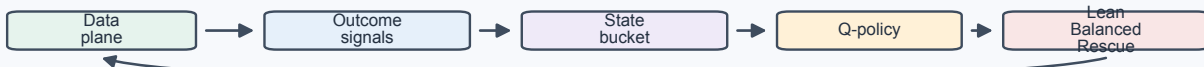
Unicast is counted as delivered only after the source receives ACK.

b Node-local online decision



$$Q \leftarrow Q + \alpha [r + \gamma \max Q' - Q]$$

c Profile selector closes the loop



profile choice rewrites route lifetime, discovery window and fallback scope

Controller state is local and small: no global topology service or neural policy is required.